0.1 Supremum, Infimum, Limit Supremum, and Limit Infimum

0.1.1 Supremum

Definition 0.1.1. Let $(X, \leq)$ be a totally ordered set, $A \subseteq X$ be a subset.
An element $\alpha \in X$ is called *Supremum* of A if:

1. $\forall s \in A,\ \ s \leq \alpha$
2. If $\beta \in X$ satisfies $\forall s \in A,\ \ s \leq \beta$, then $\alpha \leq \beta$.

Lemma 0.1.2. Let $A \subseteq \mathbb{R}$ be a subset. TFAE:

a) $s = \sup A$

b) s is an upper bound of A, and for any $\varepsilon > 0$, there exists a $r \in A$ s.t. $s - \varepsilon < r \leq s$

Proposition 0.1.3. Let $A, B \subseteq \mathbb{R}$ be non-empty subsets.

1. $\sup(A + B) = \sup A + \sup B, \quad \sup(AB) = \sup A \cdot \sup B$ if $A, B \subseteq (0, \infty)$
2. $\sup A = -\inf(-A)$
3. $\sup A = (\inf A^{-1})^{-1}$ if: $A \subseteq (0, \infty)$.
4. $\sup(A \cup B) = \max(\sup A, \sup B), \quad \sup(A \cap B) = \min(\sup A, \sup B)$

Proof. Let $\varepsilon > 0$ be given. Then, there exist $a \in A$ and $b \in B$ such that

$$\sup A + \sup B - 2\varepsilon < a + b \leq \sup A + \sup B.$$

Since $a + b$ lies in $A + B$, and clearly $\sup A + \sup B$ is an upper bound of $A + B$, thus $\sup A + \sup B = \sup(A + B)$.
To show 2), again, let $\varepsilon > 0$ be given. First, $-\sup A$ is lower bound of $-A$, because

$$\forall a \in A,\ \ a \leq \sup A \iff \forall a \in A,\ \ -a \geq -\sup A.$$

And, there exists $a \in A$ s.t.

$$\sup A - \varepsilon < a \leq \sup A.$$

This equals to

$$-\sup A \leq -a < -\sup A + \varepsilon.$$

Thus $-\sup A = \inf(-A)$.
Show 3. At first, $(\sup A)^{-1}$ is a lower bound of A^{-1}, because

$$\forall a \in A^{-1},\ \ a^{-1} \leq \sup A \iff \forall a \in A^{-1},\ \ a \geq (\sup A)^{-1}$$

That is, $(\sup A)^{-1} \leq \inf A^{-1}$. Similarly, $(\inf A^{-1})^{-1}$ is an upper bound of A. Thus, we obtain the result. $\square$

0.1.2 Supremum with functions

Definition 0.1.4. Let $f : X \to \mathbb{R}$ be a function. Denote for $A \subset X$,

$$\sup_A f \overset{\text{def}}{=} \sup\{f(x) \mid x \in A\} \text{ and } \inf_A f \overset{\text{def}}{=} \inf\{f(x) \mid x \in A\}$$

Proposition 0.1.5. Let $f, g, h : X \to \mathbb{R}$ be functions on a set X, and $A \subset X$.

1. If $f \leq g$ on A, then $\sup\limits_A f \leq \sup\limits_A g$ and $\inf\limits_A f \leq \inf\limits_A g$.
2. If $c > 0$, then $\sup\limits_A cf = c \sup\limits_A f$ and $\inf\limits_A cf = c \inf\limits_A f$.
3. If $c < 0$, then $\sup\limits_A cf = c \inf\limits_A f$ and $\inf\limits_A cf = c \sup\limits_A f$.
4. $\sup\limits_A (f+g) \leq \sup\limits_A f + \sup\limits_A g$ and $\inf\limits_A (f+g) \geq \inf\limits_A f + \inf\limits_A g$.
5. $\left|\sup\limits_A f - \sup\limits_A g\right| \leq \sup\limits_A |f-g|$ and $\left|\inf\limits_A f - \inf\limits_A g\right| \leq \sup\limits_A |f-g|$.

0.1.3 Limit Supremum

Definition 0.1.6. Let $\{a_n\}$ be a sequence of Real Numbers.
Define *limit supremum* and *limit infimum* as:

$$\limsup_{n\to\infty} a_n \overset{\text{def}}{=} \lim_{n\to\infty} \sup_{k\geq n} a_k \overset{(*)}{=} \inf_{n\in\mathbb{N}} \sup_{k\geq n} a_k \quad \text{(if } \{a_n\} \text{ not bounded above, then define as } +\infty\text{)}$$

$$\liminf_{n\to\infty} a_n \overset{\text{def}}{=} \lim_{n\to\infty} \inf_{k\geq n} a_k \overset{(*)}{=} \sup_{n\in\mathbb{N}} \inf_{k\geq n} a_k \quad \text{(if } \{a_n\} \text{ not bounded below, then define as } -\infty\text{)}$$

Note: The useful fact $(*)$ is given by the following two Lemmas:

Lemma 0.1.7. A sequence $\left\{\sup_{k\geq n} a_k\right\}_{n=1}^{\infty}$ is a monotonically decreasing sequence.

Proof. It is clear, because $n_1 < n_2 \implies \{a_k \mid k \geq n_1\} \supseteq \{a_k \mid k \geq n_2\}$. □

Lemma 0.1.8. If $\{b_n\}$ is monotonically increasing, then $\lim_{n\to\infty} b_n = \sup_{n\in\mathbb{N}} b_n$.

Proof. If $\{b_n\}$ is not bounded above, there is nothing to prove.
Assume $\{b_n\}$ is bounded above. Let $\varepsilon > 0$ be given. Then, there exists $N \in \mathbb{N}$ s.t.

$$\sup_{n\in\mathbb{N}} b_n - \varepsilon < b_N \leq \sup_{n\in\mathbb{N}} b_n$$

Since $\{b_n\}$ is increasing sequence, for all $k \geq N$,

$$\sup_{n\in\mathbb{N}} b_n - \varepsilon < b_N \leq b_k.$$

Meanwhile, by definition, for all $k \in \mathbb{N}$,

$$b_k \leq \sup_{n\in\mathbb{N}} b_n < \sup_{n\in\mathbb{N}} b_n + \varepsilon.$$

In sum, $k \geq N \implies \left|\sup_{n\in\mathbb{N}} b_n - b_k\right| < \varepsilon$. □

Note: The limit supremum is independent of the first finitely many terms of the sequence.

Lemma 0.1.9. Let $\{a_n\}$ be a sequence in $\mathbb{R}$ with bounded above.
$\alpha = \limsup_{n\to\infty} a_n$ *if and only if* it satisfies the following two conditions:

1. $\forall \varepsilon > 0,\ \exists N \in \mathbb{N}$ s.t. $n \geq N \implies a_n < \alpha + \varepsilon$ (All but finitely many terms lie below $\alpha + \varepsilon$)
2. $\forall \varepsilon > 0, \forall n \in \mathbb{N}, \exists k \geq n$ s.t. $\alpha - \varepsilon < a_k$ (Infinitely many terms lie above $\alpha - \varepsilon$)

In fact, the condition 1. implies $\limsup_{n\to\infty} a_n \leq \alpha$ and the condition 2. implies $\alpha \leq \limsup_{n\to\infty} a_n$.

Proof. $\alpha = \limsup_{n\to\infty} a_n$ if and only if: for any $\varepsilon > 0$, there exists $N \in \mathbb{N}$ such that

$$n \geq N \implies \left|\sup_{k\geq n} a_k - \alpha\right| < \varepsilon$$

That is, for $n \geq N$,

$$\left|\sup_{k\geq n} a_k - \alpha\right| < \varepsilon \iff \alpha - \varepsilon \overset{(1)}{<} \sup_{k\geq n} a_k \overset{(2)}{<} \alpha + \varepsilon$$

The inequality (2) implies the condition 1. because: $a_n \leq \sup_{k\geq n} a_k$.

The inequality (1) implies the condition 2. because: if there exists $n \in \mathbb{N}$ such that $k \geq n \implies \alpha - \varepsilon \geq a_k$, then this contradicts the inequality (1), because $\forall k \geq n,\ \alpha - \varepsilon \geq a_k \implies \alpha - \varepsilon \geq \sup_{k\geq n} a_k$.

Conversely, suppose the condition 1. Then, for any $\varepsilon > 0$, there exists $N \in \mathbb{N}$ such that

$$n \geq N \implies a_n < \alpha + \varepsilon \implies \sup_{k\geq n} a_k \leq \alpha + \varepsilon$$

Since $\varepsilon > 0$ is chosen arbitrarily,

$$\limsup_{n\to\infty} a_n = \inf_{n\in\mathbb{N}} \sup_{k\geq n} a_k \leq \alpha$$

And, suppose the condition 2. For any $\varepsilon > 0$ and $n \in \mathbb{N}$, there exists $k \geq n$ such that $\alpha - \varepsilon < a_k$, then

$$\alpha - \varepsilon < \sup_{k\geq n} a_k \implies \alpha - \varepsilon \leq \inf_{n\in\mathbb{N}} \sup_{k\geq n} a_k = \limsup_{n\to\infty} a_n$$

Similarly, since $\varepsilon > 0$ is chosen arbitrarily, we obtain the result. □

Note: The condition 2. can be stated in another way: there exist infinitely many $k \in \mathbb{N}$ such that $\alpha - \varepsilon < a_k$.

Corollary 0.1.10. Let $\{a_n\}$ be a real numbers sequence. TFAE:

a) $\alpha = \lim_{n\to\infty} a_n$.

b) $\alpha = \limsup_{n\to\infty} a_n = \liminf_{n\to\infty} a_n$.

Theorem 0.1.11. Let $\{a_n\}, \{b_n\}$, and $\{c_n\}$ be real number sequences. Then,

1. If for all $n \in \mathbb{N},\ a_n \le b_n$, then $\limsup_{n\to\infty} a_n \le \limsup_{n\to\infty} b_n$.
2. $\limsup_{n\to\infty}(a_n + b_n) \le \limsup_{n\to\infty} a_n + \limsup_{n\to\infty} b_n$
3. If for all $n \in \mathbb{N},\ a_n, b_n \ge 0$, then $\limsup_{n\to\infty}(a_n b_n) \le \left(\limsup_{n\to\infty} a_n\right) \cdot \left(\limsup_{n\to\infty} b_n\right)$
4. $\limsup_{n\to\infty} a_n = -\liminf_{n\to\infty}(-a_n)$
5. If for all $n \in \mathbb{N},\ a_n > 0$, then $\limsup_{n\to\infty} a_n = (\liminf_{n\to\infty} a_n^{-1})^{-1}$
6. If for all $n \in \mathbb{N},\ c_n > 0$, then

$$\liminf_{n\to\infty} \frac{c_{n+1}}{c_n} \le \liminf_{n\to\infty}(c_n)^{\frac{1}{n}} \le \limsup_{n\to\infty}(c_n)^{\frac{1}{n}} \le \limsup_{n\to\infty} \frac{c_{n+1}}{c_n}$$

Proof. In this proof, we use the Lemmas from the previous page. Let $\varepsilon > 0$ be given.
Proof of 1. Since the Lemma, there exists $N \in \mathbb{N}$ such that

$$k \ge N \implies b_k < \limsup_{n\to\infty} b_n + \varepsilon$$

Since $a_k \le b_k$, we obtain $\limsup_{n\to\infty} a_n \le \limsup_{n\to\infty} b_n$.
Proof of 2. Since the Lemma, there exists $N_a, N_b \in \mathbb{N}$ such that

$$k \ge N_a \implies a_k < \limsup_{n\to\infty} a_n + \varepsilon/2$$
$$k \ge N_b \implies b_k < \limsup_{n\to\infty} b_n + \varepsilon/2$$

It implies that

$$k \ge \max\{N_a, N_b\} \implies a_k + b_k < \limsup_{n\to\infty} a_n + \limsup_{n\to\infty} b_n + \varepsilon$$

Therefore we obtain the result.
Proof of 3. Similar to Proof of 2., there exists $N_a, N_b \in \mathbb{N}$ such that

$$k \ge N_a \implies a_k < \limsup_{n\to\infty} a_n + r$$
$$k \ge N_b \implies b_k < \limsup_{n\to\infty} b_n + r$$

where r is a positive root of the equation

$$x^2 + \left(\limsup_{n\to\infty} a_n + \limsup_{n\to\infty} b_n\right) x - \varepsilon = 0$$

It implies that for all $k \ge \max\{N_a, N_b\}$,

$$a_k b_k < \left(\limsup_{n\to\infty} a_n\right) \cdot \left(\limsup_{n\to\infty} b_n\right) + r\left(\limsup_{n\to\infty} a_n + \limsup_{n\to\infty} b_n\right) + r^2 < \left(\limsup_{n\to\infty} a_n\right) \cdot \left(\limsup_{n\to\infty} b_n\right) + \varepsilon$$

(The inequalities are preserved due to the positivity of the sequences.)
Proof of 4. Denote $A_n \overset{\text{def}}{=} \{a_k \mid k \ge n\}$. Since for a positive numbers set A satisfies $\sup A = -\inf(-A)$,

$$\limsup_{n\to\infty} a_n = \inf_{n\in\mathbb{N}} \sup_{k\ge n} a_k = \inf_{n\in\mathbb{N}} \sup A_n = \inf_{n\in\mathbb{N}}(-\inf(-A_n)) = -\sup_{n\in\mathbb{N}}(\inf(-A_n)) = -\sup_{n\in\mathbb{N}} \inf_{k\ge n}(-a_k) = -\liminf_{n\to\infty}(-a_n)$$

Proof of 5. Similarly Proof of 4., $\sup A = (\inf A^{-1})^{-1}$ gives the result.

Proof of 6. Enough to show the first inequality.
Put $\alpha = \liminf_{n\to\infty} \frac{c_{n+1}}{c_n}$. If $\alpha = 0$, then there is nothing to prove.
Assume $\alpha > 0$. Let $\alpha > \varepsilon > 0$ be given. Then, there exists $N \in \mathbb{N}$ such that

$$n \geq N \implies \frac{c_{n+1}}{c_n} \geq \alpha - \varepsilon$$

That is, for $n \geq N$,

$$c_n \geq (\alpha - \varepsilon)^{n-N} \cdot c_N \implies (c_n)^{\frac{1}{n}} \geq (\alpha - \varepsilon)^{1-\frac{N}{n}} \cdot c_N^{\frac{1}{n}} \to \alpha - \varepsilon$$

This implies

$$\liminf_{n\to\infty} (c_n)^{\frac{1}{n}} \geq \alpha = \liminf_{n\to\infty} \frac{c_{n+1}}{c_n}$$

$\square$

0.1.4 Subsequence

Lemma 0.1.12. The subsequential limits of a sequence $\{p_n\}$ in a Metric space (X, d) form a closed set.

Proof. Let E be the set of all subsequential limits of $\{p_n\}$, and let $q \in E'$ be a limit point of E.
If $E = \emptyset$, there is nothing to prove. Choose $n_1 \in \mathbb{N}$ such that $p_{n_1} \neq q$.
(If for all $n \in \mathbb{N},\ p_n = q$, then $E = \{q\}$, there is nothing to prove).
Suppose that $n_1, \ldots, n_{i-1}$ have been chosen. Since q is a limit point of E, there exists a point $p \in E$ such that

$$d(p, q) < \frac{1}{2^i} d(p_{n_1}, q)$$

Since $p \in E$, it is a subsequential limit of $\{p_n\}$, so there exists an index $n_i > n_{i-1}$ such that

$$d(p, p_{n_i}) < \frac{1}{2^i} d(p_{n_1}, q)$$

By the triangle inequality,

$$d(q, p_{n_i}) \leq d(p, q) + d(p, p_{n_i}) < \frac{1}{2^{i-1}} d(p_{n_1}, q)$$

This implies $d(q, p_{n_i}) \to 0 \text{ as } i \to \infty$, which means $p_{n_i} \to q$. Thus, q is a subsequential limit of $\{p_n\}$, which means $q \in E$.
Therefore, E is closed. □

Theorem 0.1.13. Let $\{a_n\}$ be a bounded sequence in $\mathbb{R}$, and $A \subseteq \mathbb{R}$ be a set of subsequential limits of $\{a_n\}$. Then,

$$\limsup_{n\to\infty} a_n = \sup A = \max A$$

Proof. Since $\{a_n\}$ is bounded, it is contained in some closed interval. Thus, it has at least one convergent subsequence.
Therefore $A \neq \emptyset$. Let $\varepsilon > 0$ be given. Then, there exists $N \in \mathbb{N}$ such that

$$k \geq N \implies a_k < \limsup_{n\to\infty} a_n + \varepsilon$$

This implies $\forall a \in A,\ a \leq \limsup_{n\to\infty} a_n + \varepsilon$, which means $\sup A \leq \limsup_{n\to\infty} a_n$.
And, there exist infinitely many terms of a_n such that

$$\limsup_{n\to\infty} a_n - \varepsilon < a_n < \limsup_{n\to\infty} a_n + \varepsilon$$

Thus there is a convergent subsequence in $\left[\limsup_{n\to\infty} a_n - \frac{\varepsilon}{2}, \limsup_{n\to\infty} a_n + \frac{\varepsilon}{2}\right]$, let p be the limit of this subsequence.

$$\limsup_{n\to\infty} a_n - \frac{\varepsilon}{2} < p \leq \sup A$$

gives the reverse inequality, thus proof complete. □

Comment for 0.1.12.
If E is finite, then closed, because $\mathbb{R}$ is T_1.
So, we can choose $p_{n_1} \in E$ s.t. $p_{n_1} \notin q$, — ILTX [illegible]
26, 07, 24.